

Matching Estimators to Censoring Rate Improves Inference with Outcome-Dependent Right-Censored Covariates

COMPSTAT 2026 | ATHENS, GREECE
August 27, 2026

Jesus E. Vazquez, PhD | JHU Provost Postdoctoral Fellow



- National Institute of Neurological Disorders and Stroke (R01NS131225)
- National Institute of Environmental Health Sciences (P30ES010126, T32E2S007018)



Tanya P. Garcia, Ph.D.



Karen Marder, M.D.

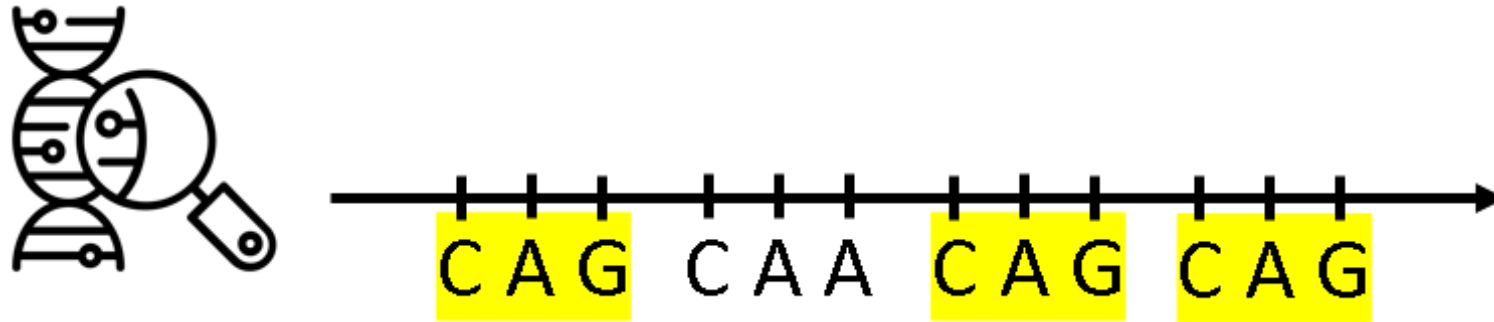


Yanyuan Ma, Ph.D.

What is Huntington disease?

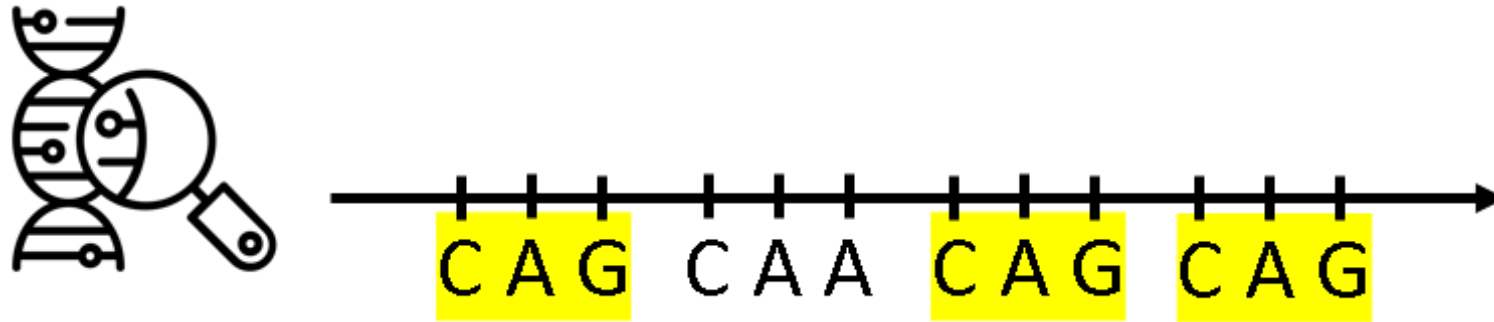
What is Huntington disease?

- Inherited neurodegenerative disease caused by a triplet repeat expansion of cytosine-adenine-guanine (CAG) in the HTT gene¹.



What is Huntington disease?

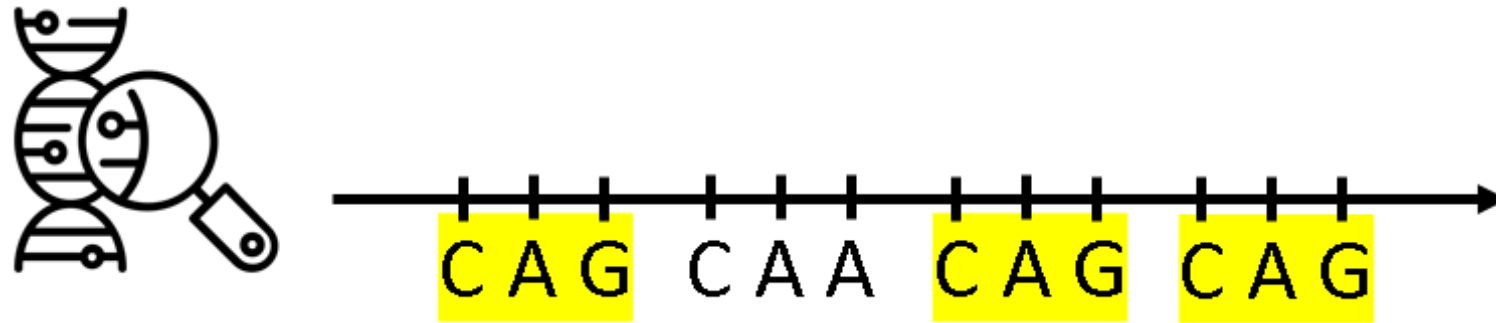
- Inherited neurodegenerative disease caused by a triplet repeat expansion of cytosine-adenine-guanine (CAG) in the HTT gene¹.



- Any individual with ≥ 40 CAG repeats will develop HD with 100% certainty².

What is Huntington disease?

- Inherited neurodegenerative disease caused by a triplet repeat expansion of cytosine-adenine-guanine (CAG) in the HTT gene¹.



- Any individual with ≥ 40 CAG repeats will develop HD with 100% certainty².

Why study Huntington disease?

- No therapies that stop or delay onset of symptoms exist².
- Triad of symptoms: motor, cognitive, and psychiatric¹.

Cognitive health

Stroop word reading test

- Assessment that requires participants to read the printed card color name while the ink is a different color.
- Score reflects the number of correct responses
- Higher indicates better cognitive health

Cognitive health

Stroop word reading test

- Assessment that requires participants to read the printed card color name while the ink is a different color.
- Score reflects the number of correct responses
- Higher indicates better cognitive health

RED

BLUE

YELLOW

GREEN

PINK

BLACK

Cognitive health

Stroop word reading test

- Assessment that requires participants to read the printed card color name while the ink is a different color.
- Score reflects the number of correct responses
- Higher indicates better cognitive health

PREDICT-HD Study

- Non-interventional multi-centered longitudinal study of Huntington disease.
- N=1,033 individuals with ≥ 40 CAG repeats.

RED

BLUE

YELLOW

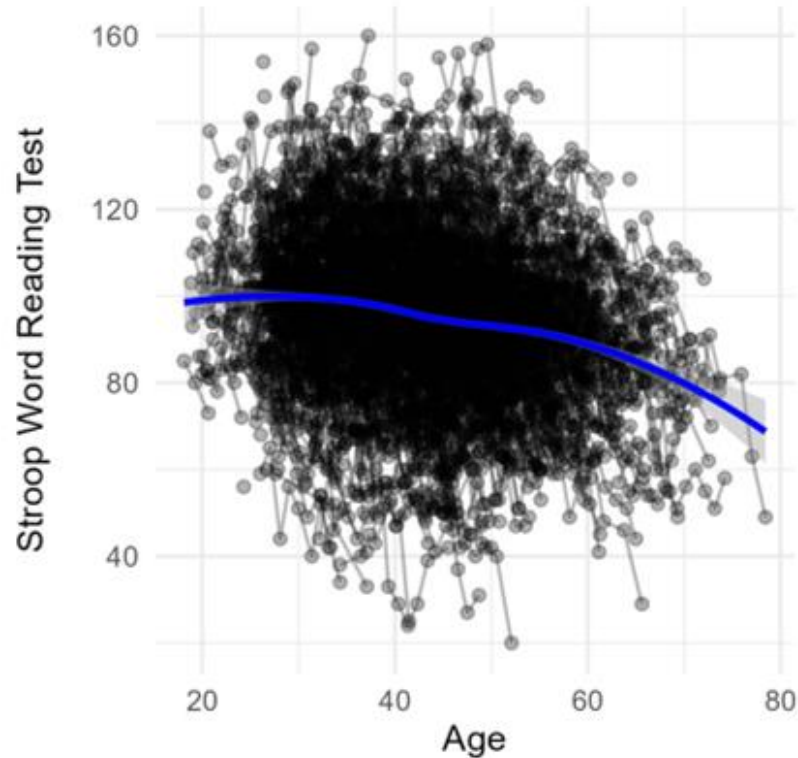
GREEN

PINK

BLACK

Cognitive health $\sim$ Time (which metric)

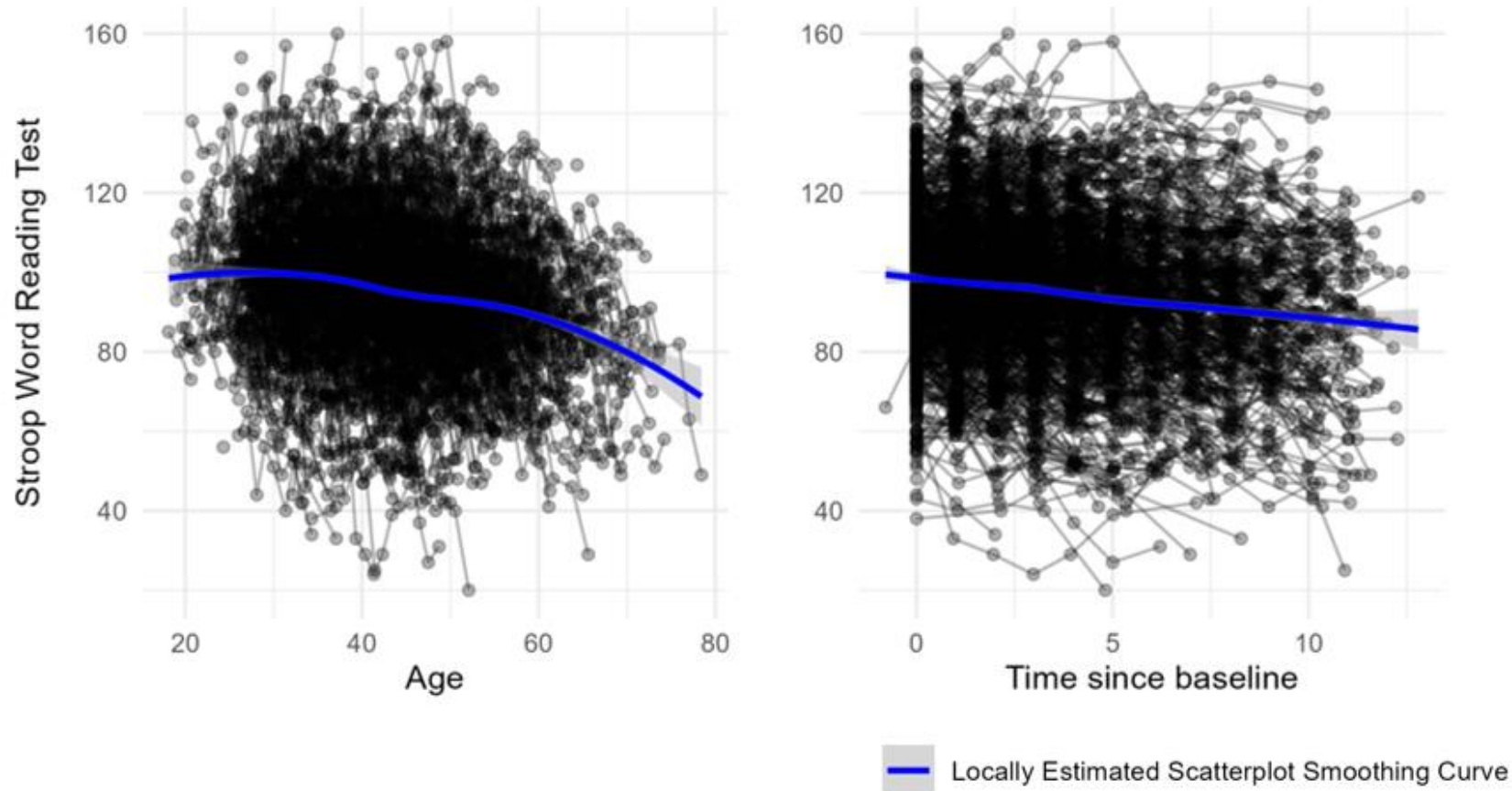
Figure 2: Stroop word reading test



— Locally Estimated Scatterplot Smoothing Curve

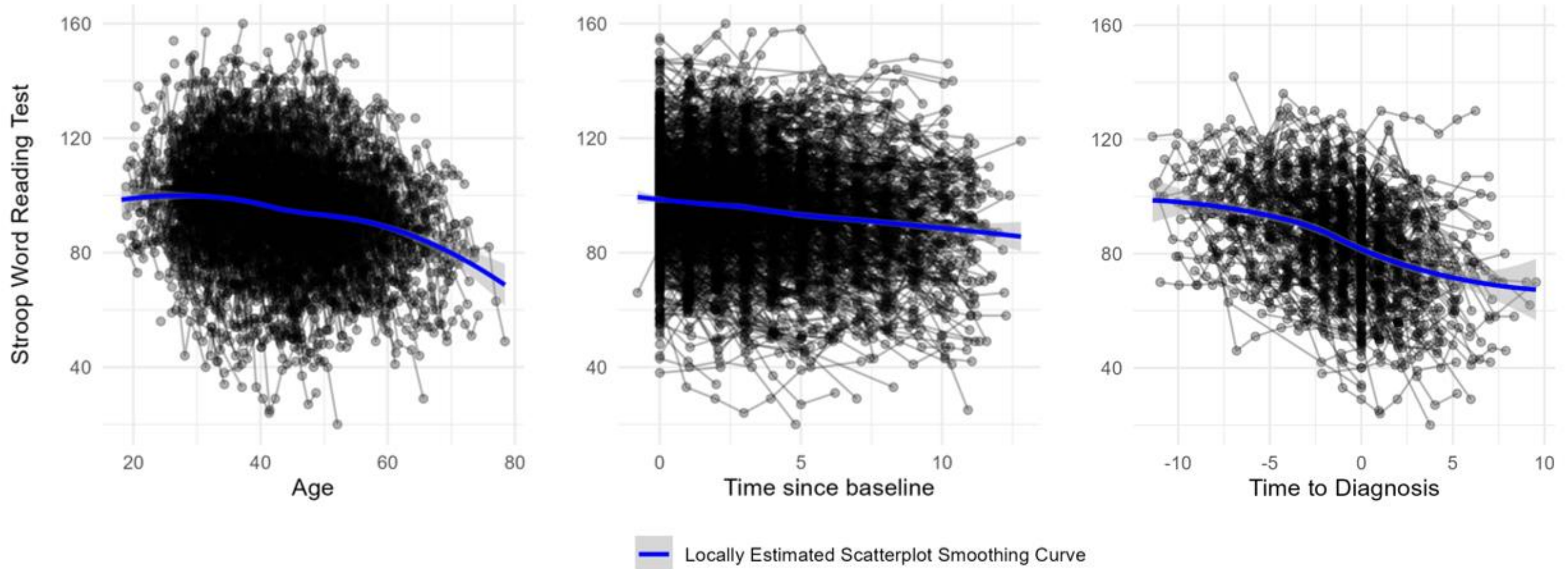
Cognitive health $\sim$ Time (which metric)

Figure 2: Stroop word reading test

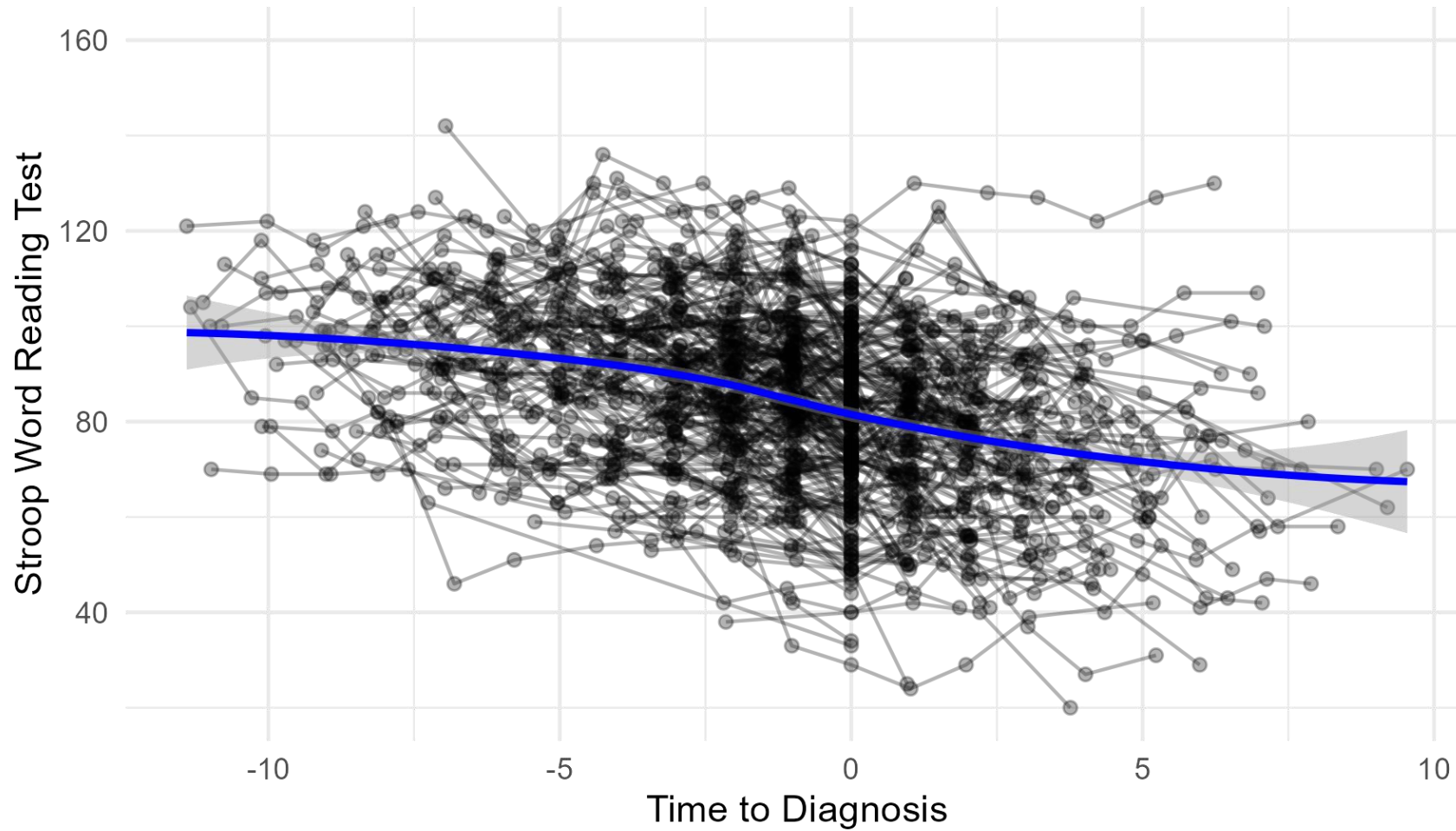


Cognitive health $\sim$ Time (which metric)

Figure 2: Stroop word reading test

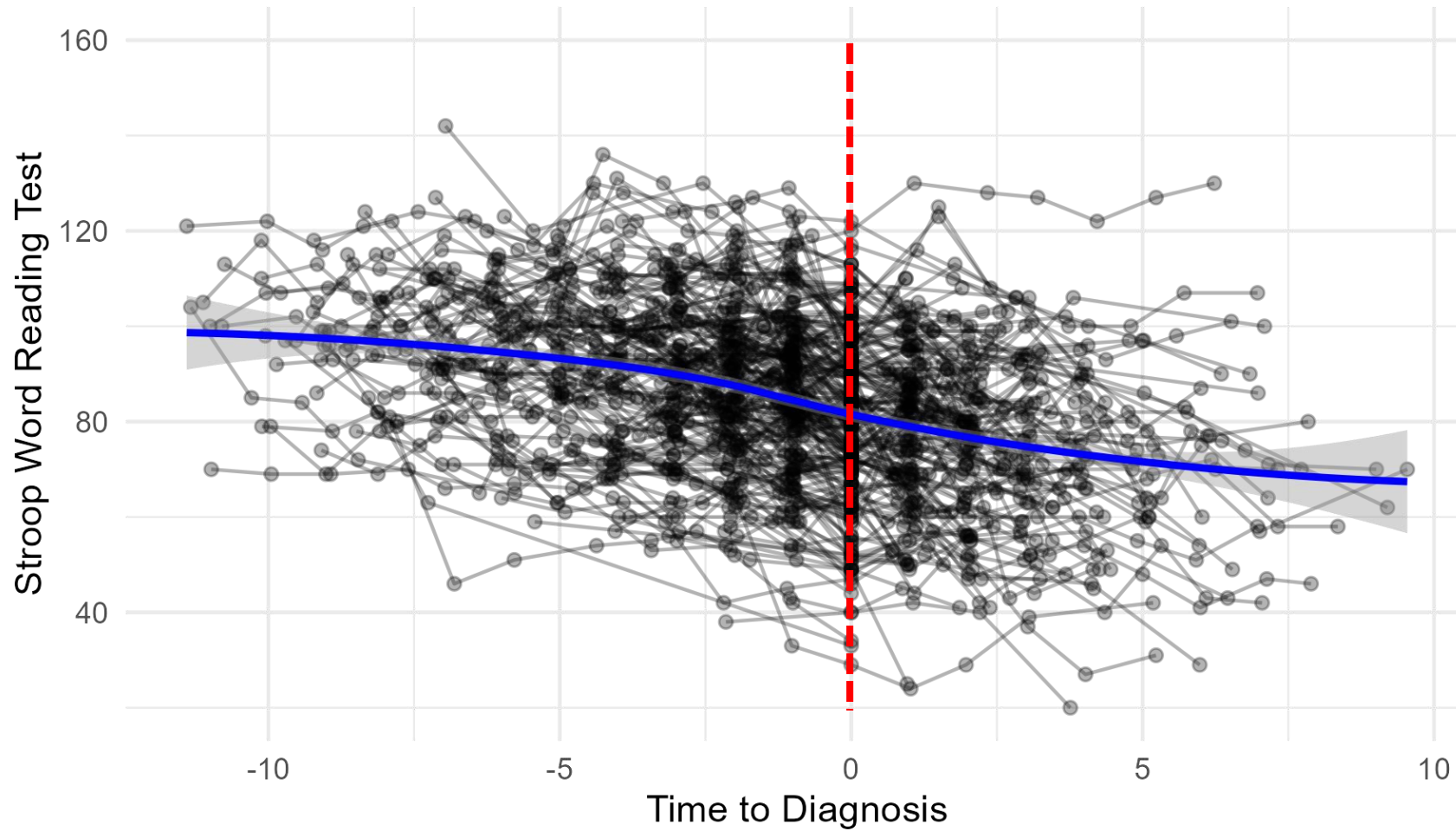


Cognitive health $\sim$ Time to Diagnosis



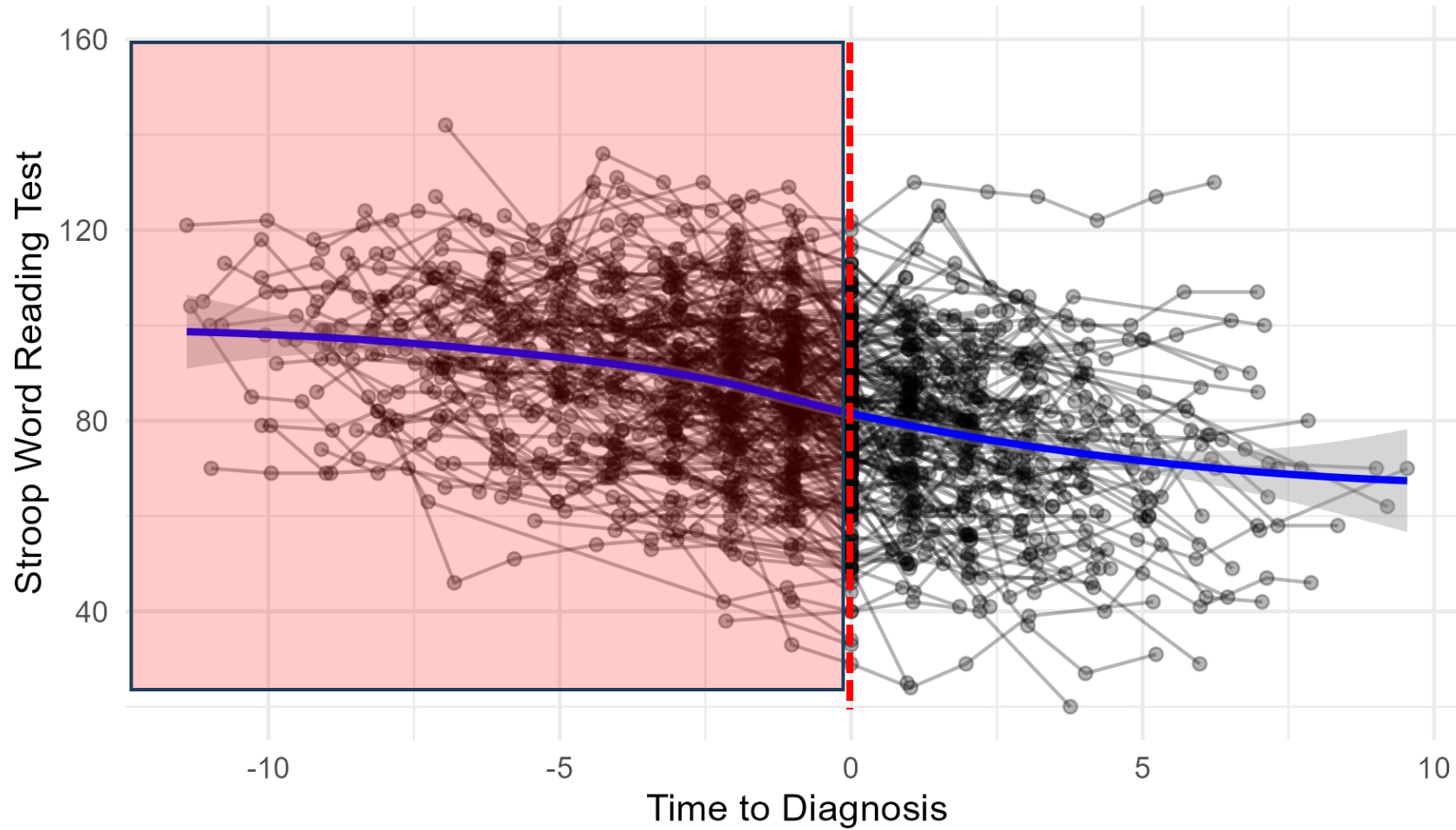
— Locally Estimated Scatterplot Smoothing Curve

Cognitive health $\sim$ Time to Diagnosis



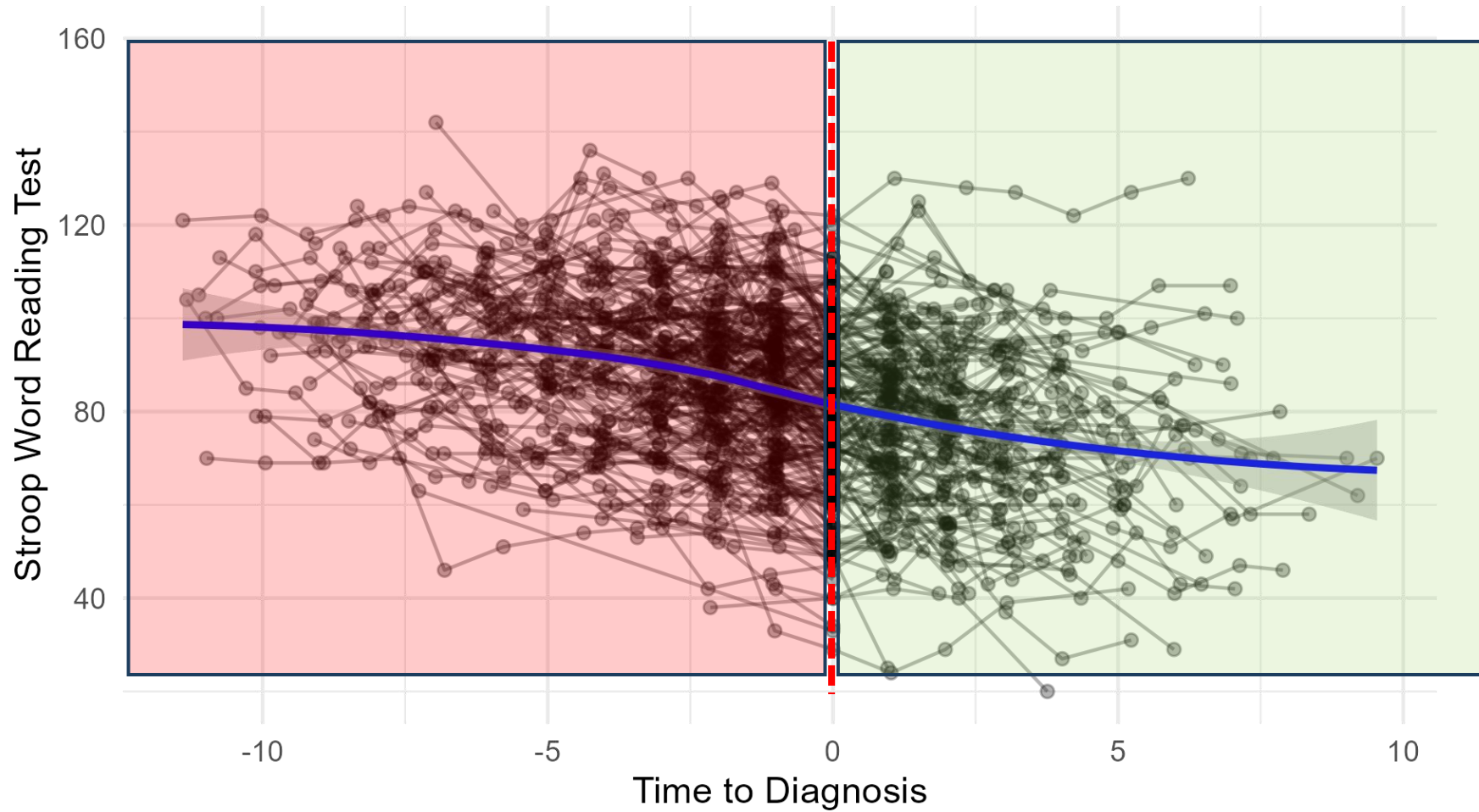
— Locally Estimated Scatterplot Smoothing Curve

Cognitive health $\sim$ Time to Diagnosis



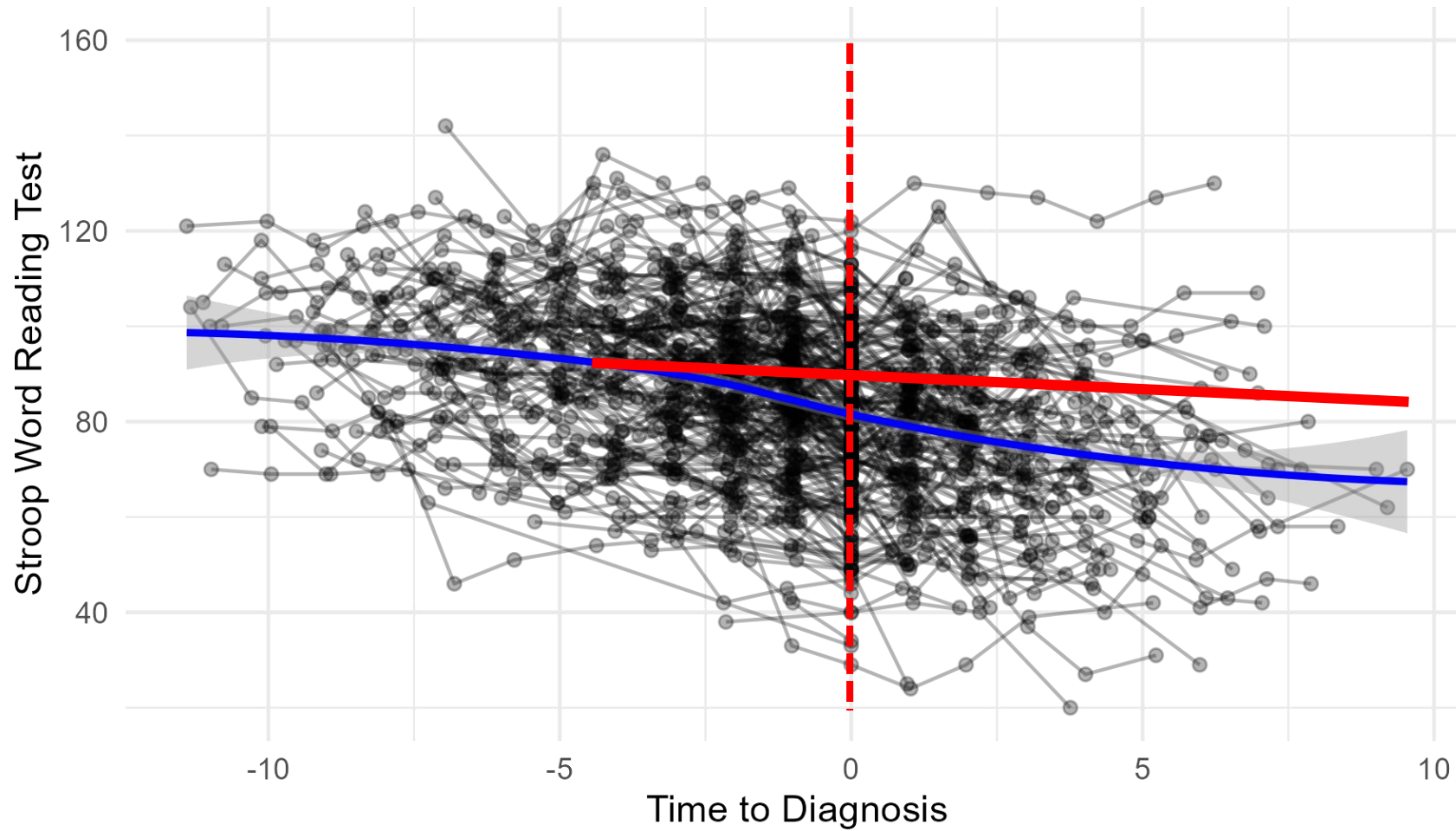
— Locally Estimated Scatterplot Smoothing Curve

Cognitive health $\sim$ Time to Diagnosis



— Locally Estimated Scatterplot Smoothing Curve

Cognitive health $\sim$ Time to Diagnosis



— Locally Estimated Scatterplot Smoothing Curve

What is the statistical problem?



Bejamin button process (Revival analysis)

References the idea of analyzing time in reverse, leading up to an event of interest³.



Bejamin button process in action

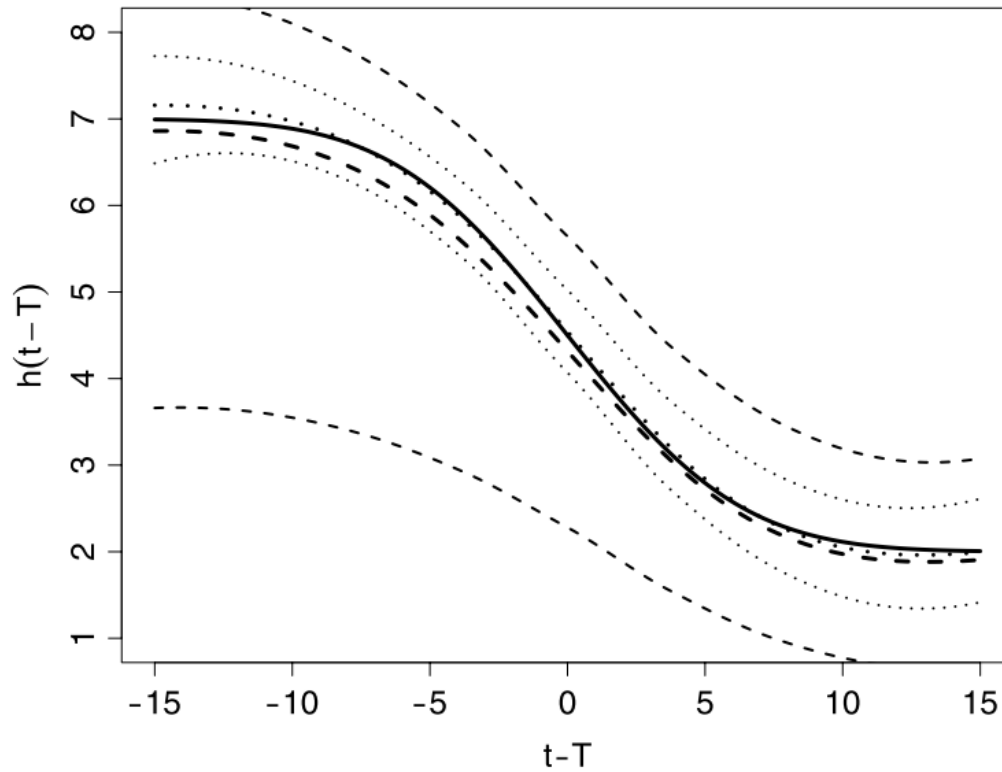


Figure 3: Hormonal levels (2010)⁴

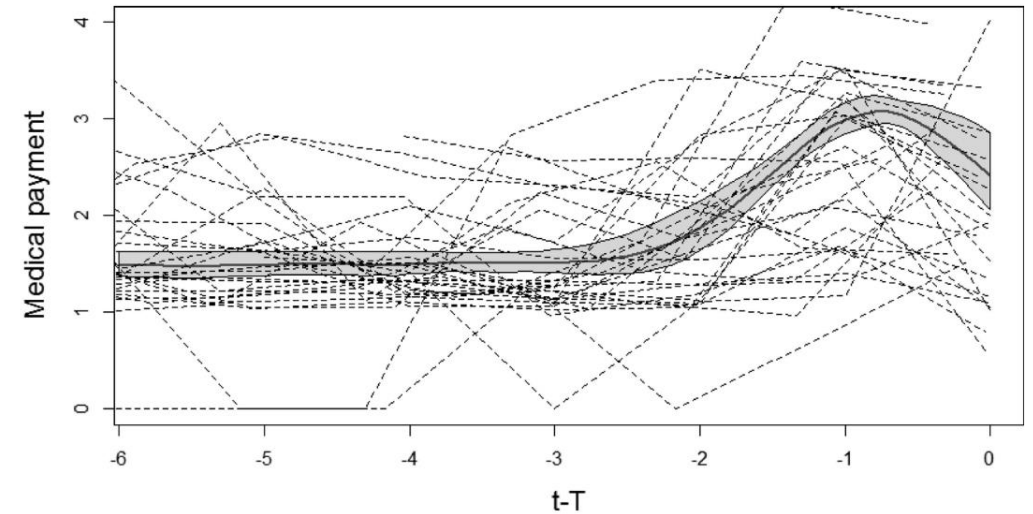


Figure 4: Medical Payments (2018)⁵

Bejamin button process in action, cont.

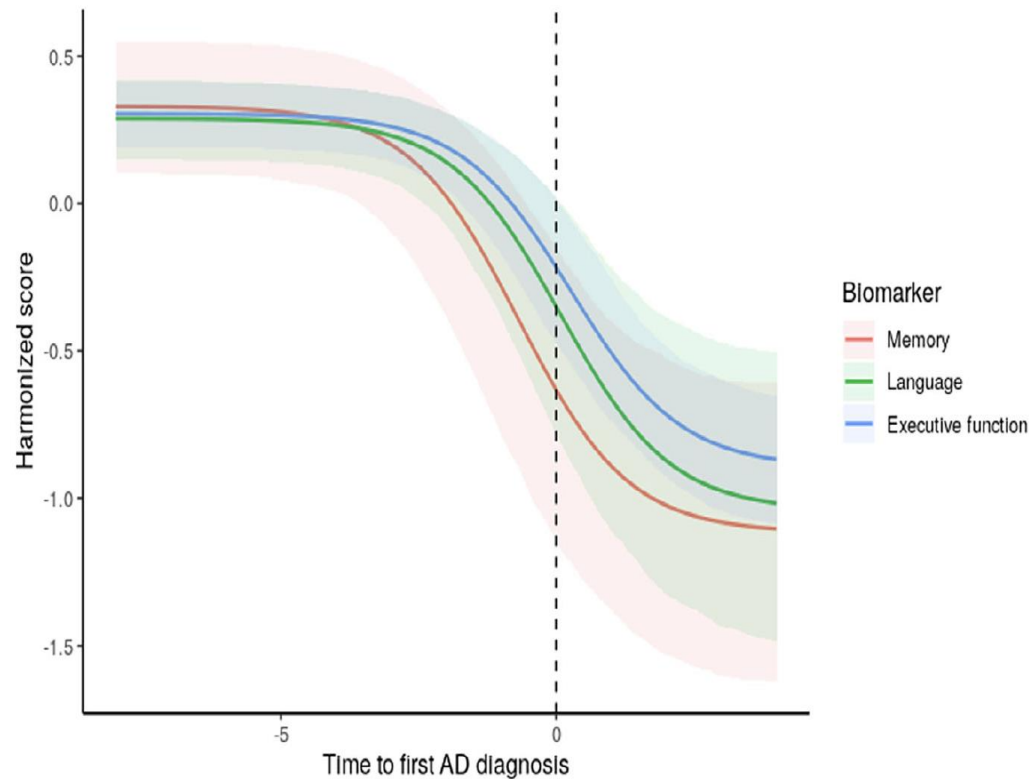


Figure 5: Cognitive markers in Alzheimer's disease (2024)⁶

Deep interest in how health markers and medical payments evolve around a clinical milestone (e.g., menarche, death, or motor diagnosis).

This information is useful in clinical care, management, and the development of clinical trials aimed at developing treatments.

Right-censored covariate problem

Often called revival analysis, deals with the problem of analyzing time in reverse, leading up to an event of interest. For the i^{th} subject, consider the regression model

$$Y_i = \mathbf{m}(X_i, \mathbf{Z}_i; \boldsymbol{\beta}) + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$



Right-censored covariate problem

Often called revival analysis, deals with the problem of analyzing time in reverse, leading up to an event of interest. For the i^{th} subject, consider the regression model

$$Y_i = \mathbf{m}(X_i, \mathbf{Z}_i; \boldsymbol{\beta}) + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$

Statistical Problem: Instead of observing X we observe (W, Δ)



Right-censored covariate problem

Often called revival analysis, deals with the problem of analyzing time in reverse, leading up to an event of interest. For the i^{th} subject, consider the regression model

$$Y_i = \mathbf{m}(X_i, \mathbf{Z}_i; \boldsymbol{\beta}) + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$$

Statistical Problem: Instead of observing X we observe (W, Δ)

We are interested in estimating $\boldsymbol{\theta} = (\boldsymbol{\beta}^T, \sigma)$ using (Y, W, Δ, Z) , where (Y, Z) are always observed but $W = \min(X, C)$ and $\Delta = I(X \leq C)$.



Right-censored covariate problem

Q1: Can a right-censored covariate be treated as a missing covariate?



Right-censored covariate problem

Q1: Can a right-censored covariate be treated as a missing covariate?

When X is missing, the observed data is (Y, XR, R, Z) , whereas the observed data for the right-censored covariate is (Y, W, Δ, Z) .

Answer: Depends. In many settings you can introduce bias⁷.



Right-censored covariate problem

Q1: Can a right-censored covariate be treated as a missing covariate?

When X is missing, the observed data is (Y, X_R, R, Z) , whereas the observed data for the right-censored covariate is (Y, W, Δ, Z) .

Answer: Depends. In many settings you can introduce bias⁷.

 International Statistical Review



International Statistical Review (2026) doi: 10.1111/insr.70043

How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem

Jesus E. Vazquez¹ , Marissa C. Ashner², Yanyuan Ma³, Karen Marder⁴ and Tanya P. Garcia⁵

Assumptions

Outcome independent censoring

$$Y \perp\!\!\!\perp C \mid (X, Z)$$

- Independent censoring: $X \perp\!\!\!\perp C \mid Z$
- Dependent censoring: $X \not\perp\!\!\!\perp C \mid Z$



Right-censored covariate problem

Q1: Can a right-censored covariate be treated as a missing covariate?

When X is missing, the observed data is (Y, X_R, R, Z) , whereas the observed data for the right-censored covariate is (Y, W, Δ, Z) .

Answer: Depends. In many settings you can introduce bias⁷.

 International Statistical Review



International Statistical Review (2026) doi: 10.1111/insr.70043

How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem

Jesus E. Vazquez¹ , Marissa C. Ashner², Yanyuan Ma³, Karen Marder⁴ and Tanya P. Garcia⁵

Estimators validated

- Complete case (CC)
- Inverse probability weighting (IPW)
- Maximum likelihood estimator (MLE)

Estimators proposed

- Augmented complete case (ACC)
- Modified ACC (MACC)
- Augmented IPW (AIPW)



What is outcome dependent censoring?



Outcome dependent censoring



Outcome dependent censoring

Most estimators assume outcome-independent censoring



Outcome dependent censoring

Most estimators assume outcome-independent censoring

- $C \perp\!\!\!\perp Y \mid (X, Z)$ ^{7,8,9}



Outcome dependent censoring

Most estimators assume outcome-independent censoring

- $C \perp\!\!\!\perp Y \mid (X, Z)$ ^{7,8,9}
- $C \perp\!\!\!\perp (Y, X) \mid Z$ ^{10,11}



Outcome dependent censoring

Most estimators assume outcome-independent censoring

- $C \perp\!\!\!\perp Y \mid (X, Z)$ ^{7,8,9}
- $C \perp\!\!\!\perp (Y, X) \mid Z$ ^{10,11}

This assumption is reasonable when censoring arises from administrative factors such as predetermined study end dates. However, in many studies of progressive diseases, censoring depends directly on the outcome, violating outcome-independent assumptions.



Outcome dependent censoring

Outcome influences censoring

Individuals with high cUHDRS scores (better health) may not be diagnosed during the study period. Therefore:

- $C \perp\!\!\!\perp Y \mid (X, Z)$ does not hold $\rightarrow C \not\perp\!\!\!\perp Y \mid (X, Z)$.
- Estimators that rely on this assumption will lead to bias.



Outcome dependent censoring

Outcome influences censoring

Individuals with high cUHDRS scores (better health) may not be diagnosed during the study period. Therefore:

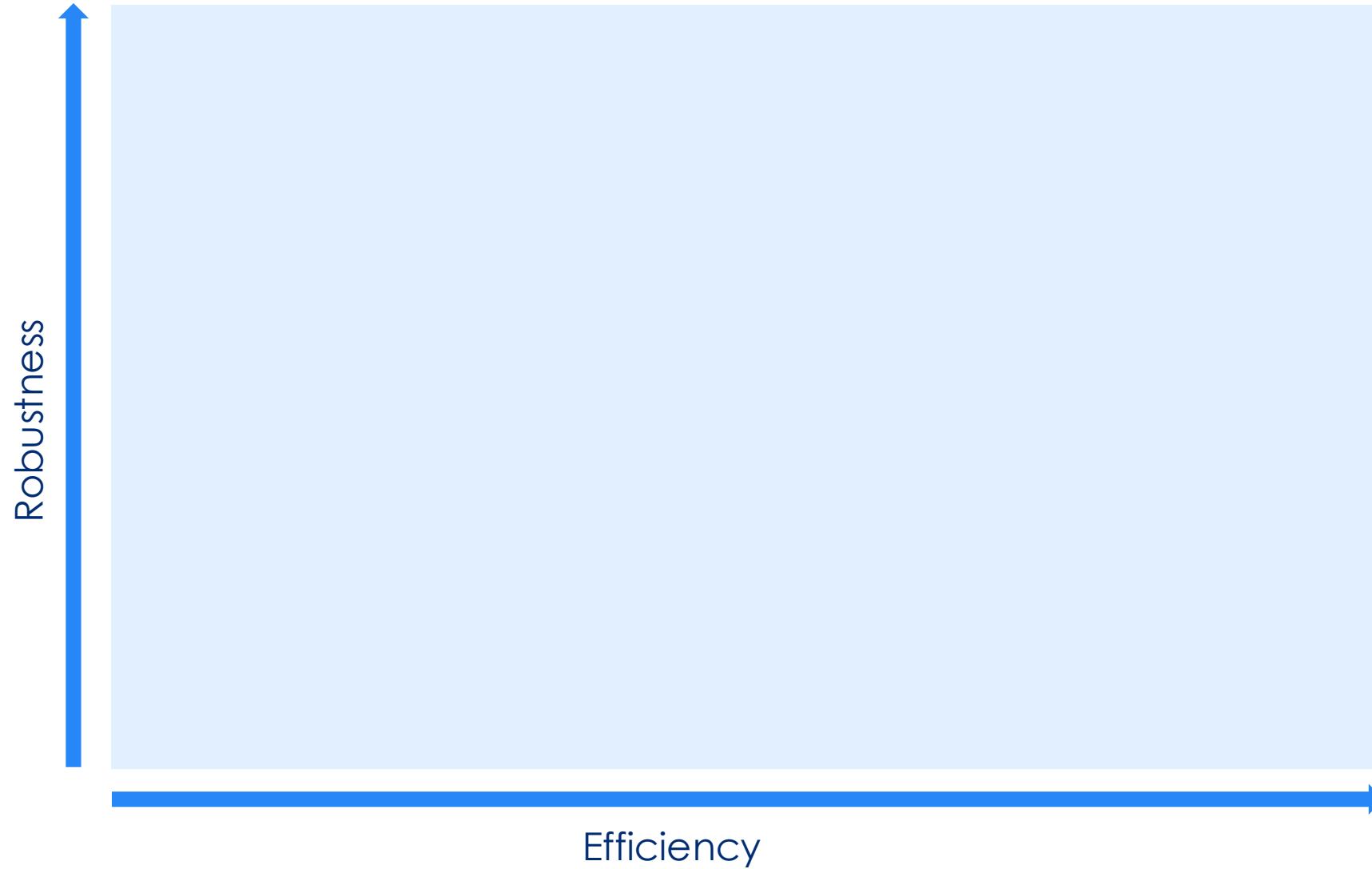
- $C \perp\!\!\!\perp Y \mid (X, Z)$ does not hold $\rightarrow C \not\perp\!\!\!\perp Y \mid (X, Z)$.
- Estimators that rely on this assumption will lead to bias.

Important for investigating cUHDRS

Need to develop consistent estimators that may estimate the parameters indexing the regression model:

$$cUHDRS_i = \beta_0 + \beta_1 TTMCI_i + \beta_2 CAG_i + \beta_3 Education_i + \beta_4 Sex_i + \epsilon_i$$

Our goal



Our goal



Eye bird view of contributions

Consistent Estimators

- IPW estimator^{13,14}.
- AIPW estimator and improved form.
- Maximum likelihood estimator.



Eye bird view of contributions

Consistent Estimators

- IPW estimator^{13,14}.
- AIPW estimator and improved form.
- Maximum likelihood estimator.

Inconsistent Estimators

- Complete case estimator.
- ACC estimator and improved forms.
- MACC and improved forms.
- Conditional mean imputation estimator.

Inverse probability weighting (IPW)

Probability of complete observation

$$\begin{aligned}\pi_{Y,X,Z}(y, w, \mathbf{z}) &= \Pr(C > w | Y = y, X = x, \mathbf{Z} = \mathbf{z}) \\ &= \int I(C \geq w) f_{C|Y,X,\mathbf{Z}}(c, y, x, \mathbf{z}) dc \\ &= \int I(C \geq w) f_{C|Y,\mathbf{Z}}(c, y, \mathbf{z}) dc\end{aligned}$$

Inverse probability weighting (IPW)

Probability of complete observation

$$\begin{aligned}\pi_{Y,X,Z}(y, w, \mathbf{z}) &= \Pr(C > w | Y = y, X = x, \mathbf{Z} = \mathbf{z}) \\ &= \int I(C \geq w) f_{C|Y,X,\mathbf{Z}}(c, y, x, \mathbf{z}) dc \\ &= \int I(C \geq w) f_{C|Y,\mathbf{Z}}(c, y, \mathbf{z}) dc\end{aligned}$$

Assumptions

1. $C \perp\!\!\!\perp Y | (X, Z)$
2. $C \perp\!\!\!\perp X | (Y, Z)$



Inverse probability weighting (IPW)

Probability of complete observation

$$\begin{aligned}\pi_{Y,X,Z}(y, w, \mathbf{z}) &= \Pr(C > w | Y = y, X = x, \mathbf{Z} = \mathbf{z}) \\ &= \int I(C \geq w) f_{C|Y,X,Z}(c, y, x, \mathbf{z}) dc \\ &= \int I(C \geq w) f_{C|Y,Z}(c, y, \mathbf{z}) dc\end{aligned}$$

Assumptions

1. $C \perp\!\!\!\perp Y | (X, Z)$
2. $C \perp\!\!\!\perp X | (Y, Z)$

Estimating Equations

$$\sum_{i=1}^n \frac{\delta_i \mathbf{S}_{\boldsymbol{\theta}}^F(y_i, w_i, \mathbf{z}_i)}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)} = \mathbf{0}, \text{ where } \mathbf{S}_{\boldsymbol{\theta}}^F(y_i, x_i, \mathbf{z}_i) = \frac{\partial}{\partial \boldsymbol{\theta}^T} \log f_{Y|X,Z}(y_i, x_i, \mathbf{z}_i; \boldsymbol{\theta})$$

Augmented Inverse probability weighting (AIPW)

Augments the IPW estimator to improve efficiency.

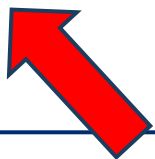
Estimating Equations

$$\sum_{i=1}^n \left[\underbrace{\frac{\delta_i \mathbf{S}_{\boldsymbol{\theta}}^F(y_i, w_i, \mathbf{z}_i)}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)}}_{\text{IPW estimator}} + \underbrace{\left\{ 1 - \frac{\delta_i}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)} \right\} \Psi_{\text{AIPW}}(y_i, \mathbf{z}_i; \boldsymbol{\theta})}_{\text{Augmentation term}} \right] = \mathbf{0}$$

Augmented Inverse probability weighting (AIPW)

Augments the IPW estimator to improve efficiency.

Estimating Equations

$$\sum_{i=1}^n \left[\underbrace{\frac{\delta_i \mathbf{S}_{\theta}^F(y_i, w_i, \mathbf{z}_i)}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)}}_{\text{IPW estimator}} + \underbrace{\left\{ 1 - \frac{\delta_i}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)} \right\} \Psi_{\text{AIPW}}(y_i, \mathbf{z}_i; \theta)}_{\text{Augmentation term}} \right] = 0$$


Form of augmentation term (efficient choice)

$$\Psi_{\text{AIPW}}(y_i, \mathbf{z}_i; \theta) = \frac{E_{X|Y,Z}[\{1 - 1/\pi_{Y,X,Z}(y_i, X, \mathbf{z}_i)\} \mathbf{S}_{\theta}^F(y_i, X, \mathbf{z}_i)]}{E_{X|Y,Z}\{1 - 1/\pi_{Y,X,Z}(y_i, X, \mathbf{z}_i)\}}$$

Improved AIPW

Modifies the AIPW estimator to guarantee improved efficiency.

Estimating Equations

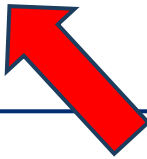
$$\sum_{i=1}^n \left[\underbrace{\frac{\delta_i \mathbf{S}_{\boldsymbol{\theta}}^F(y_i, w_i, \mathbf{z}_i)}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)}}_{\text{IPW estimator}} + \underbrace{\left\{ 1 - \frac{\delta_i}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)} \right\} \boldsymbol{\Lambda}_{\text{AIPW}} \Psi_{\text{AIPW}}(y_i, \mathbf{z}_i; \boldsymbol{\theta})}_{\text{Augmentation term}} \right] = \mathbf{0}$$



Improved AIPW

Modifies the AIPW estimator to guarantee improved efficiency.

Estimating Equations

$$\sum_{i=1}^n \left[\underbrace{\frac{\delta_i \mathbf{S}_{\boldsymbol{\theta}}^F(y_i, w_i, \mathbf{z}_i)}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)}}_{\text{IPW estimator}} + \underbrace{\left\{ 1 - \frac{\delta_i}{\pi_{Y,X,Z}(y_i, w_i, \mathbf{z}_i)} \right\} \boldsymbol{\Lambda}_{\text{AIPW}} \Psi_{\text{AIPW}}(y_i, \mathbf{z}_i; \boldsymbol{\theta})}_{\text{Augmentation term}} \right] = \mathbf{0}$$


Form of augmentation term (efficient choice)

$$\boldsymbol{\Lambda}_{\text{AIPW}} = E_{Y,W,\Delta,Z} [\{1 - \Delta/\pi_{Y,X,Z}(Y, W, \mathbf{Z})\} \mathbf{S}_{\boldsymbol{\theta}}^F(Y, W, \mathbf{Z}) \Psi_{\text{AIPW}}(Y, \mathbf{Z}; \boldsymbol{\theta})^T]$$
$$E_{Y,W,\Delta,Z} \left(\left[\{1 - \Delta/\pi_{Y,X,Z}(Y, W, \mathbf{Z})\} \Psi_{\text{AIPW}}(Y, \mathbf{Z}; \boldsymbol{\theta}) \right]^{\otimes 2} \right)^{-1}$$



Maximum likelihood estimator

The MLE estimator takes the partial derivative of log-likelihood with respect to θ^T .

Estimating Equations



$$\sum_{i=1}^n \left[\underbrace{\delta_i \mathbf{S}_{\theta}^F(y_i, w_i, \mathbf{z}_i)}_{\text{CC estimator}} + (1 - \delta_i) \underbrace{\frac{\int_{w_i < x} \mathbf{S}_{\theta}^F(y_i, x, \mathbf{z}_i) f_{Y|X,Z}(y_i, x, \mathbf{z}_i) f_{X|Z}(x, \mathbf{z}_i) dx}{\int_{w_i < x} f_{Y|X,Z}(y_i, x, \mathbf{z}_i) f_{X|Z}(x, \mathbf{z}_i) dx}}_{\text{Censored data term}} \right] = \mathbf{0}$$

Each component is not zero in expectation, but together, the expectation of the estimating equation is equal to zero.

Maximum likelihood estimator

The MLE estimator takes the partial derivative of log-likelihood with respect to θ^T .

Estimating Equations

$$\sum_{i=1}^n \left[\underbrace{\delta_i \mathbf{S}_{\theta}^F(y_i, w_i, \mathbf{z}_i)}_{\text{CC estimator}} + (1 - \delta_i) \underbrace{\frac{\int_{w_i < x} \mathbf{S}_{\theta}^F(y_i, x, \mathbf{z}_i) f_{Y|X,Z}(y_i, x, \mathbf{z}_i) f_{X|Z}(x, \mathbf{z}_i) dx}{\int_{w_i < x} f_{Y|X,Z}(y_i, x, \mathbf{z}_i) f_{X|Z}(x, \mathbf{z}_i) dx}}_{\text{Censored data term}} \right] = \mathbf{0}$$


Each component is not zero in expectation, but together, the expectation of the estimating equation is equal to zero.

Maximum likelihood estimator

The MLE estimator takes the partial derivative of log-likelihood with respect to θ^T .

Estimating Equations

$$\sum_{i=1}^n \left[\underbrace{\delta_i \mathbf{S}_{\theta}^F(y_i, w_i, \mathbf{z}_i)}_{\text{CC estimator}} + (1 - \delta_i) \underbrace{\frac{\int_{w_i < x} \mathbf{S}_{\theta}^F(y_i, x, \mathbf{z}_i) f_{Y|X,Z}(y_i, x, \mathbf{z}_i) f_{X|Z}(x, \mathbf{z}_i) dx}{\int_{w_i < x} f_{Y|X,Z}(y_i, x, \mathbf{z}_i) f_{X|Z}(x, \mathbf{z}_i) dx}}_{\text{Censored data term}} \right] = \mathbf{0}$$


Each component is not zero in expectation, but together, the expectation of the estimating equation is equal to zero.

Estimator choice depends on right-censoring rate

Consistency depends on the correct nuisance distribution specification

- IPW and AIPW (both forms): require correct $f_{C|Y,Z}(c, y, \mathbf{z})$
- MLE: requires correct $f_{X|Z}(x, \mathbf{z})$



Estimator choice depends on right-censoring rate

Consistency depends on the correct nuisance distribution specification

- IPW and AIPW (both forms): require correct $f_{C|Y,Z}(c, y, \mathbf{z})$
- MLE: requires correct $f_{X|Z}(x, \mathbf{z})$

High Censoring rates

We observe $(Y, C, \mathbf{Z})$ more often than $(Y, X, \mathbf{Z})$: easier to estimate $f_{C|Y,Z}(c, y, \mathbf{z})$.

Low Censoring rates

We observe $(Y, X, \mathbf{Z})$ more often than $(Y, C, \mathbf{Z})$: easier to estimate $f_{X|Z}(x, \mathbf{z})$.



Estimator choice depends on right-censoring rate

Consistency depends on the correct nuisance distribution specification

- IPW and AIPW (both forms): require correct $f_{C|Y,Z}(c, y, \mathbf{z})$
- MLE: requires correct $f_{X|Z}(x, \mathbf{z})$

High Censoring rates

We observe $(Y, C, \mathbf{Z})$ more often than $(Y, X, \mathbf{Z})$: easier to estimate $f_{C|Y,Z}(c, y, \mathbf{z})$.

Low Censoring rates

We observe $(Y, X, \mathbf{Z})$ more often than $(Y, C, \mathbf{Z})$: easier to estimate $f_{X|Z}(x, \mathbf{z})$.

Our recommendations:

- **High censoring rate ($\geq 70\%$): IPW or AIPW**
- **Low censoring rate ($< 70\%$): MLE**



Simulations: Bias

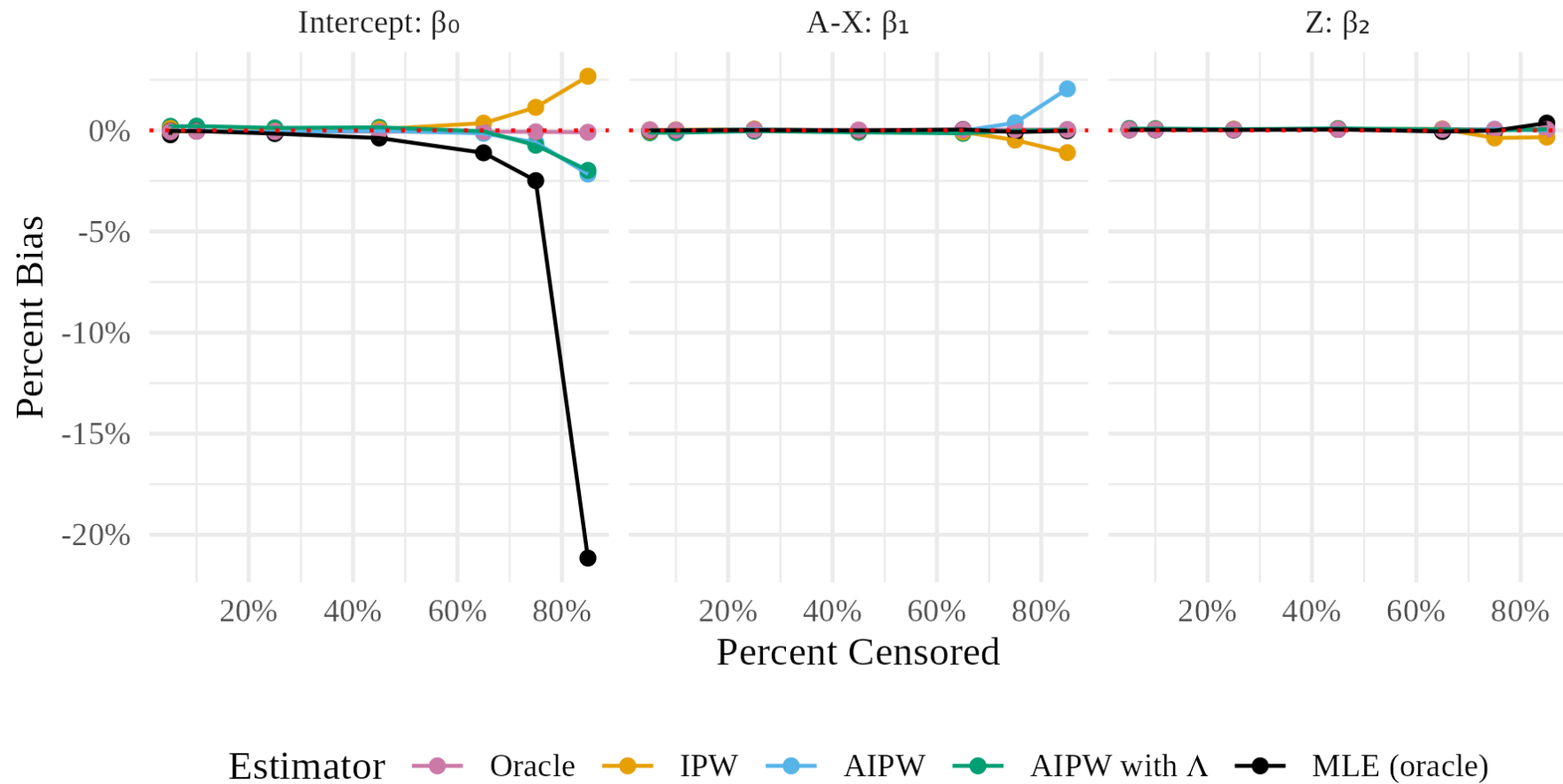


Figure 6: Bias

Simulations: Coverage

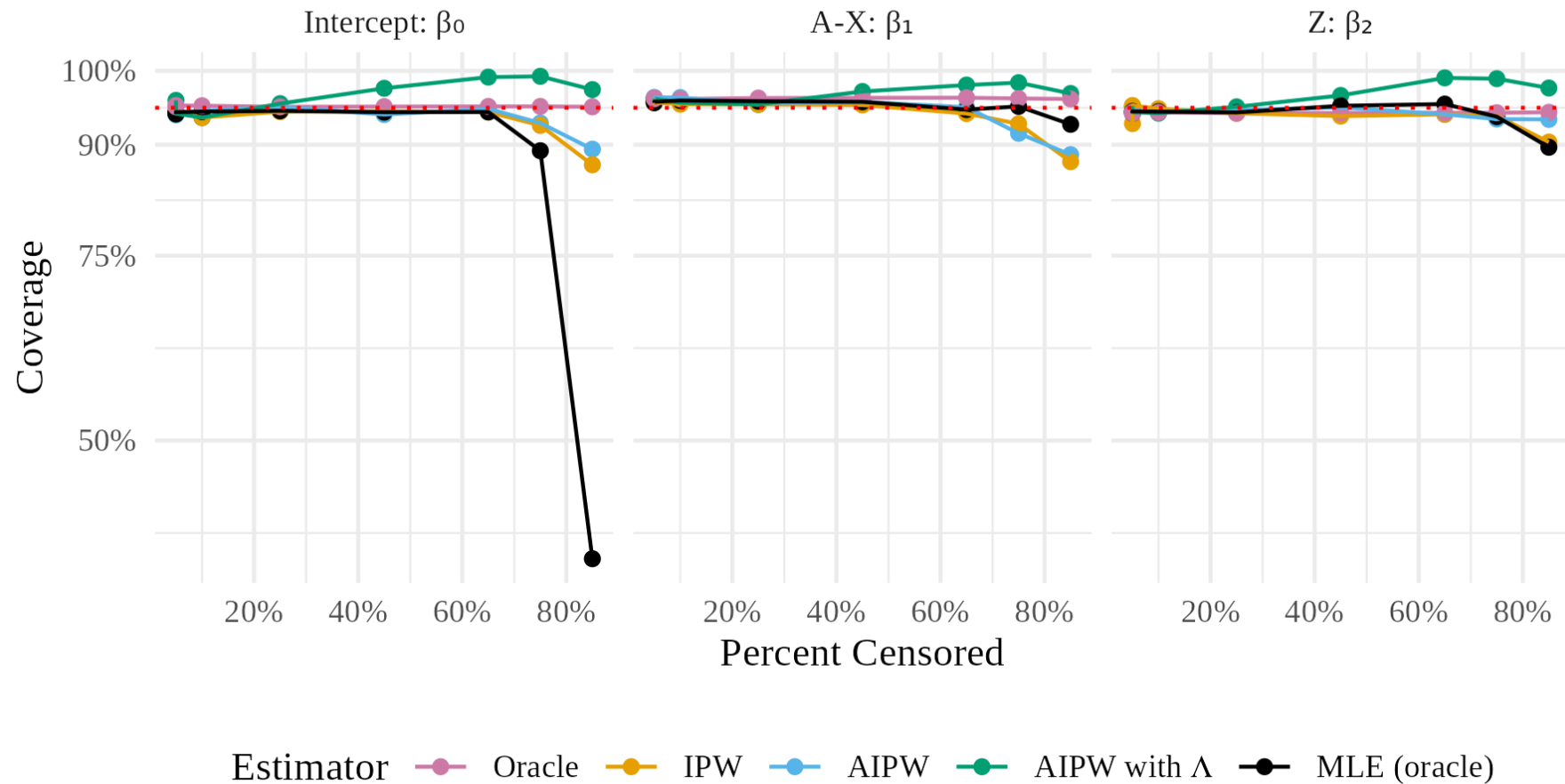


Figure 6: Coverage

Simulations: Efficiency

Estimator	Specification	Bias (%)	SE	SD	Cov
Intercept: $\beta_0 = 1$					
IPW	correct $\pi_{Y,X,Z}$	-0.21	6.13	6.37	93.61
	incorrect $\pi_{Y,X,Z}$	-21.89	6.17	6.30	6.19
	estimated $\pi_{Y,X,Z}$	-0.07	5.61	5.62	94.05
MLE	correct $f_{X Z}$	-0.06	4.03	3.99	95.41
	incorrect $f_{X Z}$	-5.86	4.01	3.97	68.45
	estimated $f_{X Z}$	0.10	4.18	4.01	95.82
AIPW	without using Λ				
	correct $\pi_{Y,X,Z}$ and correct $\Psi_{\text{AIPW}}(y, \mathbf{z})$	0.11	5.15	5.14	94.52
	incorrect $\pi_{Y,X,Z}$ and correct $\Psi_{\text{AIPW}}(y, \mathbf{z})$	-13.45	6.53	6.50	46.29
	correct $\pi_{Y,X,Z}$ and incorrect $\Psi_{\text{AIPW}}(y, \mathbf{z})$	-0.17	11.49	11.40	94.85
	incorrect $\pi_{Y,X,Z}$ and incorrect $\Psi_{\text{AIPW}}(y, \mathbf{z})$	25.50	15.29	15.66	63.41
	estimated $\pi_{Y,X,Z}$ and estimated $\Psi_{\text{AIPW}}(y, \mathbf{z})$	0.05	4.75	4.67	94.65
	using Λ				
	correct $\pi_{Y,X,Z}$	-0.63	4.99	5.11	93.95
	incorrect $\pi_{Y,X,Z}$	-16.84	5.34	5.48	12.61
	estimated $\pi_{Y,X,Z}$	-0.57	5.47	4.79	96.27



Simulations: Efficiency

Estimator	Specification	Bias	SE	SD	Cov
Intercept: $\beta_0 = 1$					
IPW	correct $\pi_{Y,X,Z}$	-0.21	6.13	6.37	93.61
	incorrect $\pi_{Y,X,Z}$	-21.89	6.17	6.30	6.19
	estimated $\pi_{Y,X,Z}$	-0.07	5.61	5.62	94.05
MLE	correct $f_{X Z}$	-0.06	4.03	3.99	95.41
	incorrect $f_{X Z}$	-5.86	4.01	3.97	68.45
	estimated $f_{X Z}$	0.10	4.18	4.01	95.82
AIPW	<u>without using Λ</u>				
	correct $\pi_{Y,X,Z}$ and correct $\Psi_{\text{AIPW}}(y, \mathbf{z})$	0.11	5.15	5.14	94.52
	incorrect $\pi_{Y,X,Z}$ and correct $\Psi_{\text{AIPW}}(y, \mathbf{z})$	-13.45	6.53	6.50	46.29
	correct $\pi_{Y,X,Z}$ and incorrect $\Psi_{\text{AIPW}}(y, \mathbf{z})$	-0.17	11.49	11.40	94.85
	incorrect $\pi_{Y,X,Z}$ and incorrect $\Psi_{\text{AIPW}}(y, \mathbf{z})$	25.50	15.29	15.66	63.41
	estimated $\pi_{Y,X,Z}$ and estimated $\Psi_{\text{AIPW}}(y, \mathbf{z})$	0.05	4.75	4.67	94.65
	<u>using Λ</u>				
	correct $\pi_{Y,X,Z}$	-0.63	4.99	5.11	93.95
	incorrect $\pi_{Y,X,Z}$	-16.84	5.34	5.48	12.61
	estimated $\pi_{Y,X,Z}$	-0.57	5.47	4.79	96.27



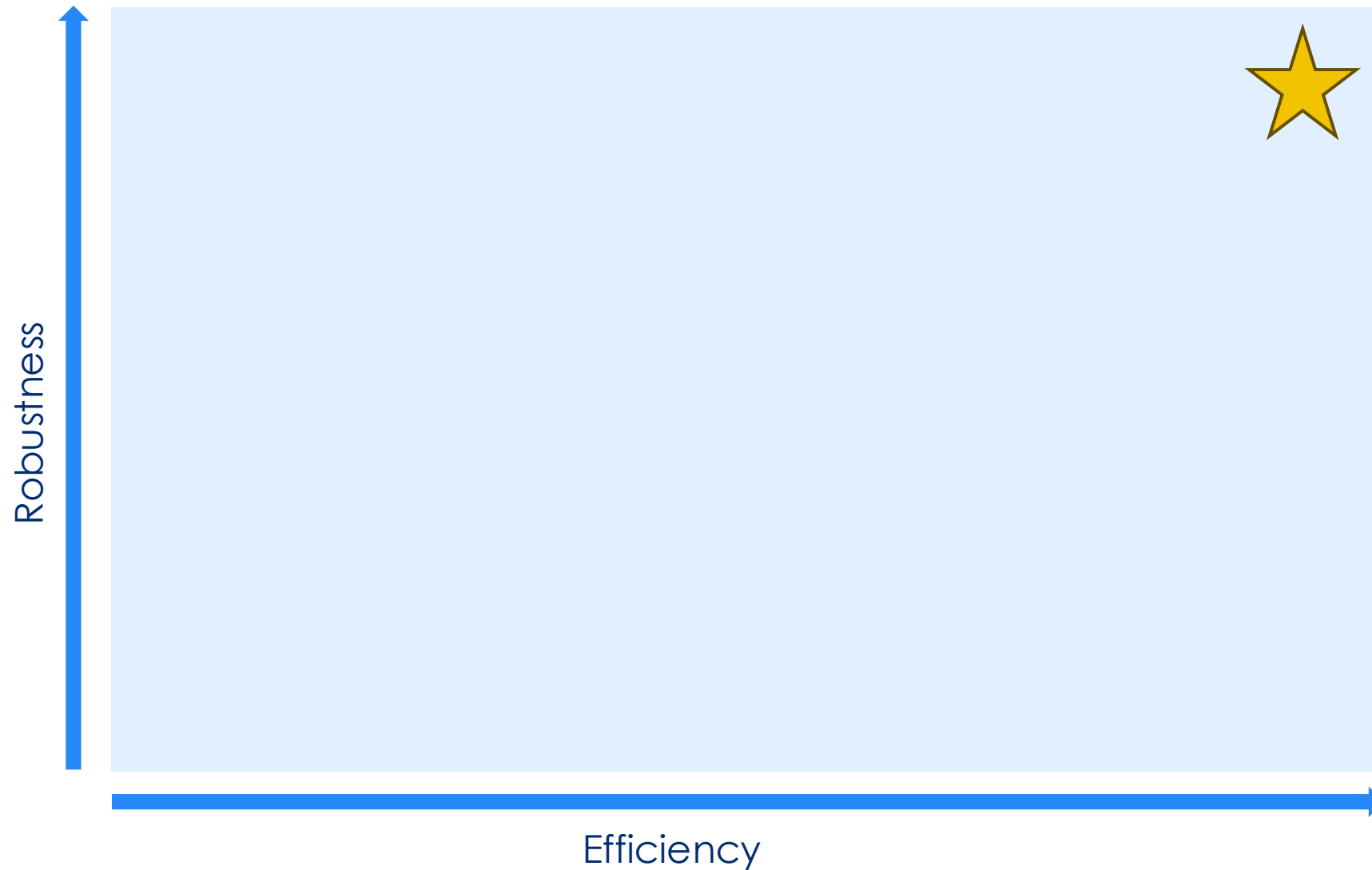
Application to Huntington disease

Table 1: Linear regression parameter estimates (standard errors) of the composite Unified Huntington Disease Rating Scale (cUHDRS), PREDICT-HD ($n = 833$).

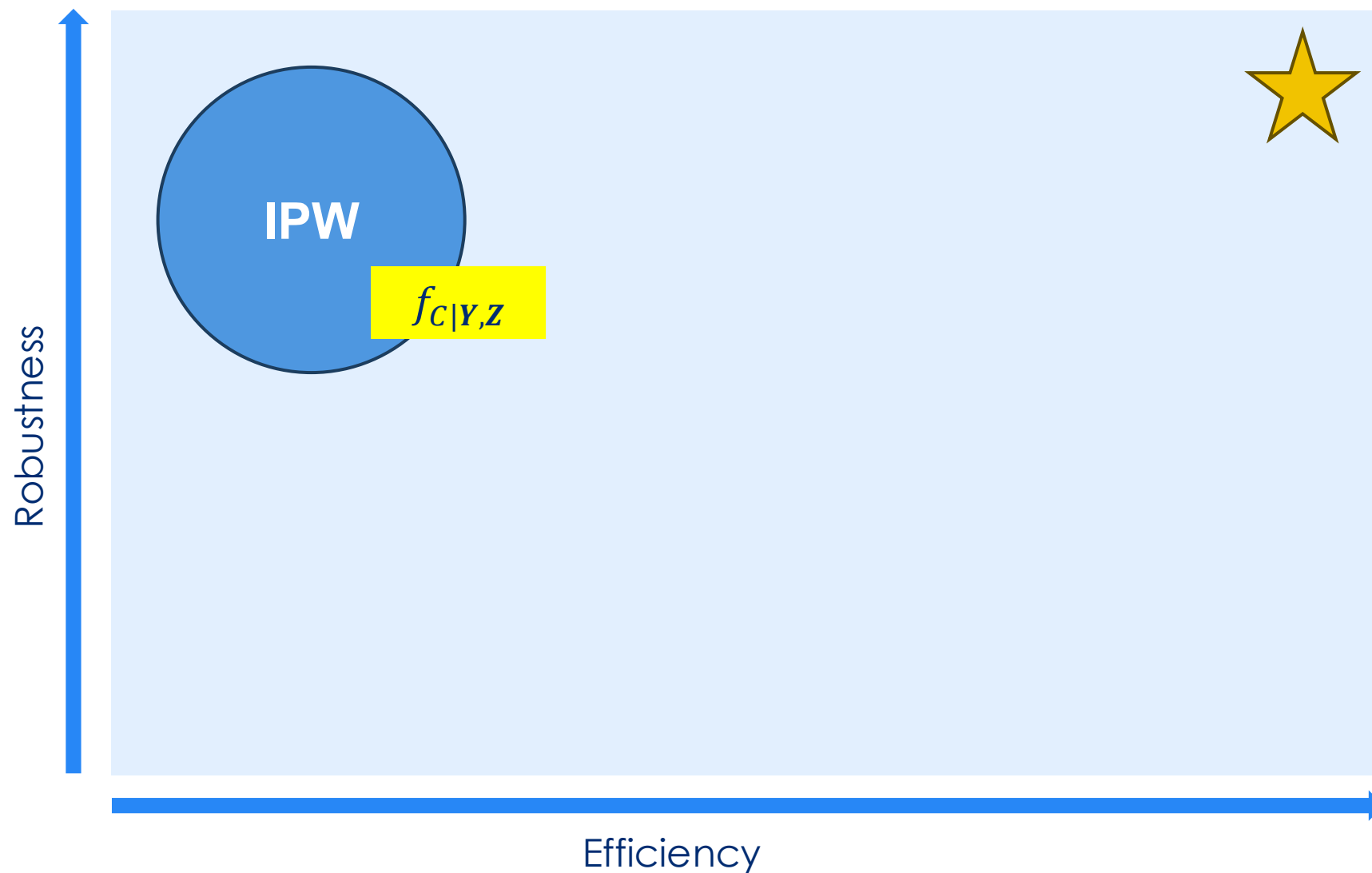
Estimator	Intercept	TTMCI (Years)	CAPs	Education (Years)	Sex (Female)
IPW	-1.927 (0.247)**	-0.166 (0.040)**	-0.022 (0.016)	0.038 (0.049)	0.295 (0.227)
AIPW	-0.991 (0.130)**	-0.034 (0.003)**	-0.042 (0.006)**	0.084 (0.019)**	0.225 (0.117)
AIPW with Λ	-1.826 (0.283)**	-0.148 (0.038)**	-0.026 (0.014)	0.047 (0.034)	0.198 (0.202)
MLE	-1.298 (0.286)**	-0.054 (0.017)**	-0.028 (0.044)	0.050 (0.006)**	0.141 (0.239)

Note: cUHDRS and Education are centered at their mean; CAP Score is centered at its mean and scaled by a factor of 10; TTMCI refers to Time to Mild Cognitive Impairment, which is calculated as current age minus age at cognitive impairment.

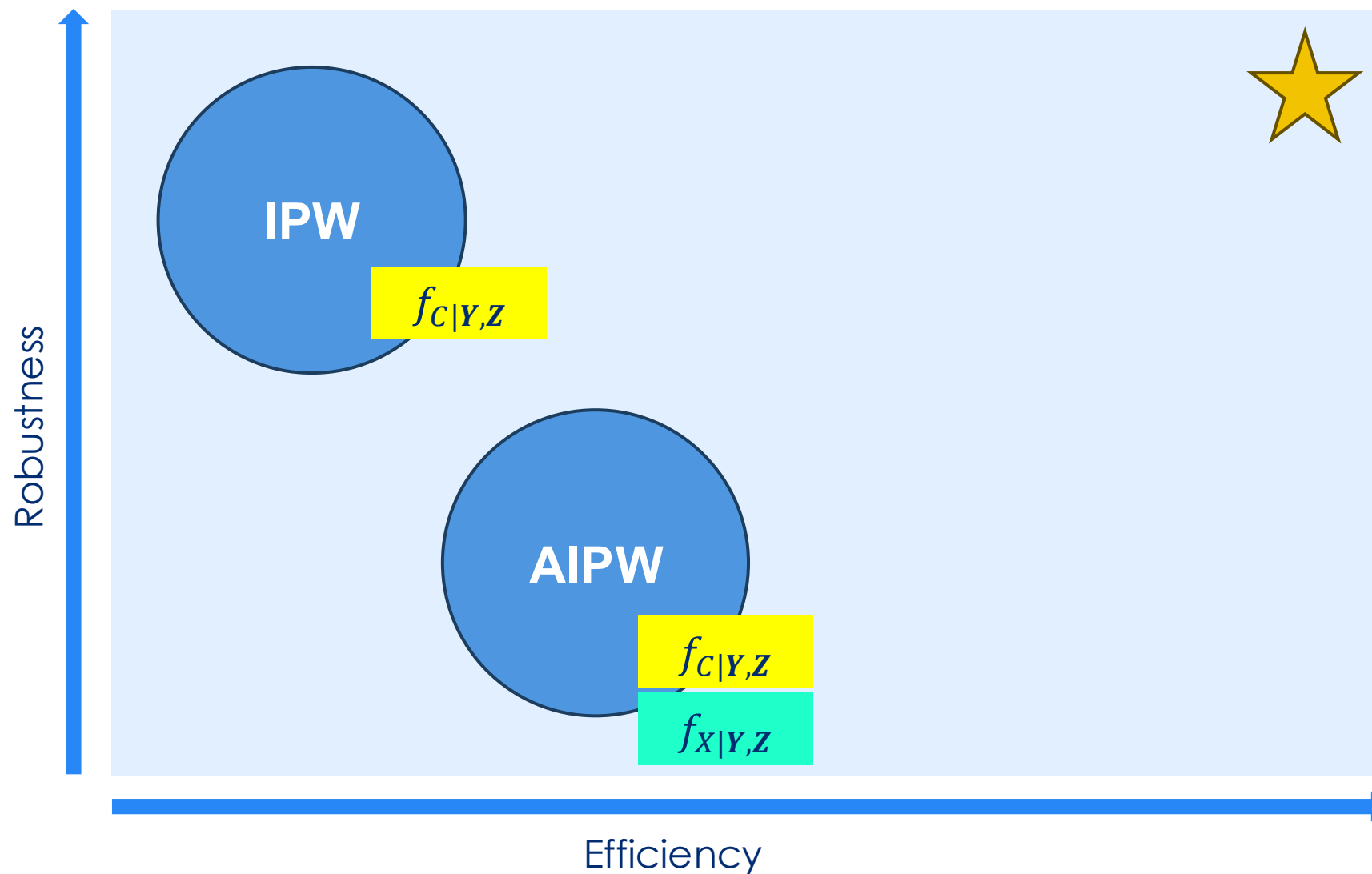
Eye bird view of contributions



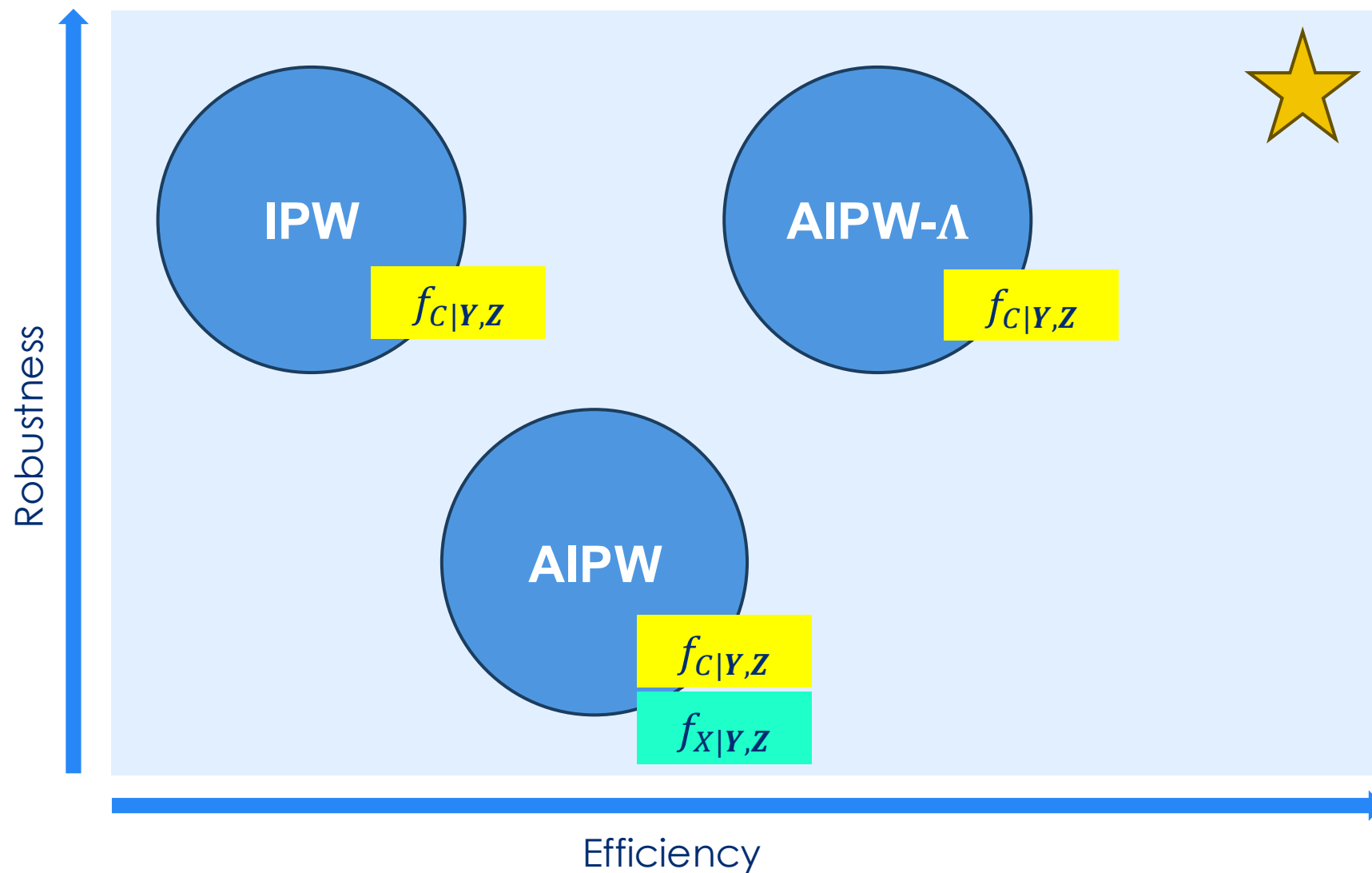
Eye bird view of contributions



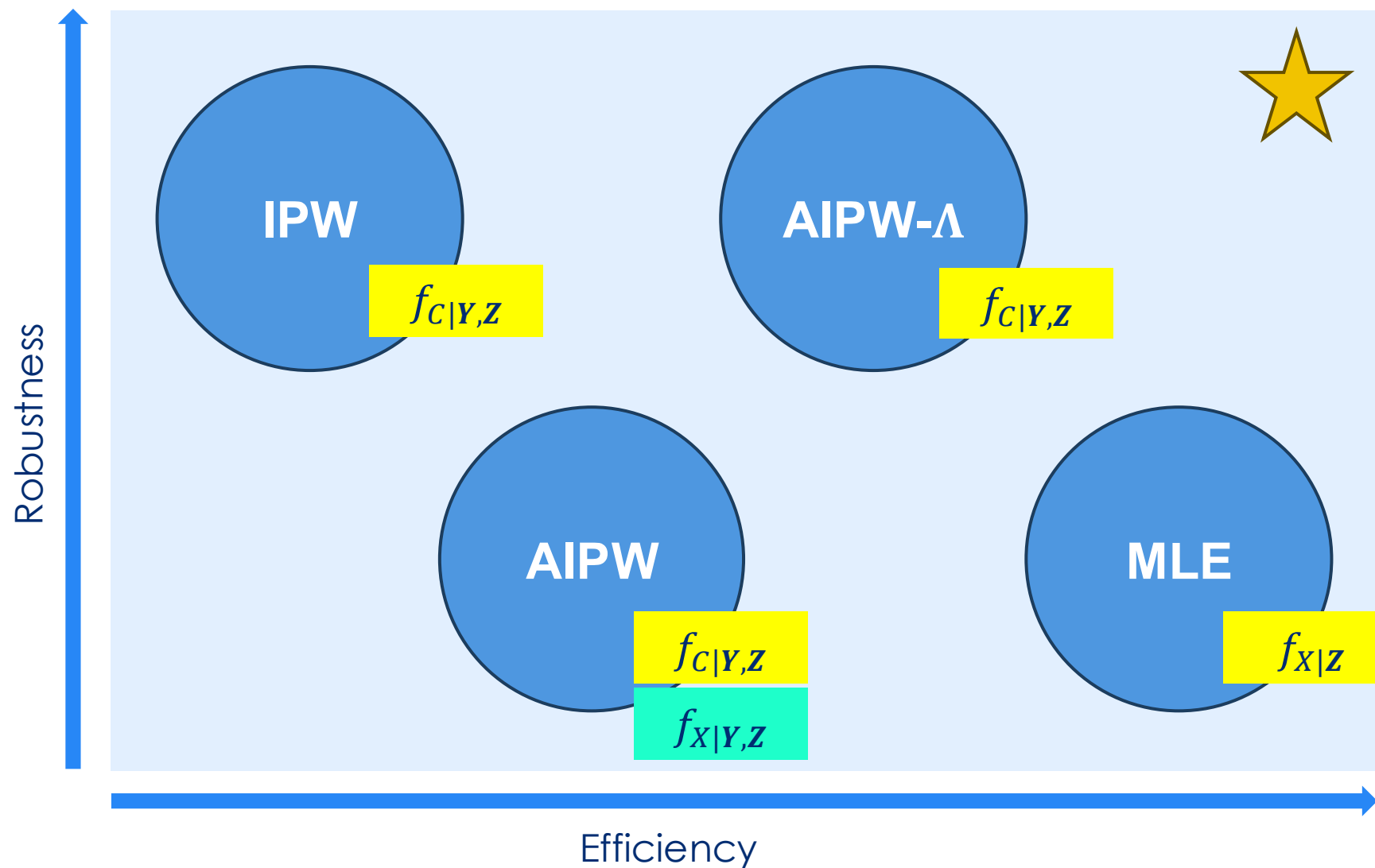
Eye bird view of contributions



Eye bird view of contributions



Eye bird view of contributions

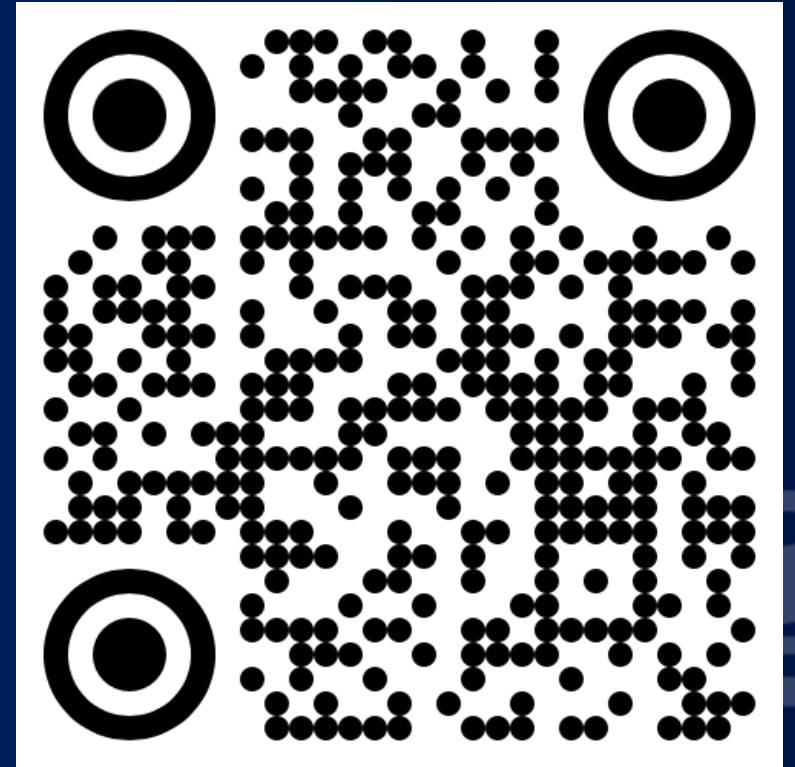


Thank You!

jvazqu18@jh.edu

Jesus E. Vazquez, Ph.D.

Provost's Postdoctoral Fellow



<https://arxiv.org/abs/2511.15929>

Works cited

1. MacDonald, Marcy E., et al. "The Huntington's Disease Collaborative Research Group. A novel gene containing a trinucleotide repeat that is expanded and unstable on Huntington's disease chromosomes." *Cell* 72.6 (1993): 971-983.
2. Tabrizi, Sarah J., et al. "Potential disease-modifying therapies for Huntington's disease: lessons learned and future opportunities." *The Lancet Neurology* 21.7 (2022): 645-658.
3. Dempsey, Walter, and Peter McCullagh. "Survival models and health sequences." *Lifetime data analysis* 24.4 (2018): 550-584.
4. Lu, Xuewen, et al. "Longitudinal data analysis with event time as a covariate." *Statistics in Biosciences* 2.1 (2010): 65-80.
5. Kong, Shengchun, et al. "Conditional modeling of longitudinal data with terminal event." *Journal of the American Statistical Association* 113.521 (2018): 357-368.
6. Kang, Kaidi, et al. "Comparing the dynamics of multiple cognitive markers for Alzheimer's disease progression using a novel double anchoring events-based sigmoidal mixed model." *Alzheimer's & Dementia* 20 (2024): e085501.
7. Vazquez, Jesus E., et al. "Establishing the Parallels and Differences Between Right-Censored and Missing Covariates." *arXiv preprint arXiv:2409.04684* (2024).
8. Ashner, Marissa C., and Tanya P. Garcia. "Understanding the implications of a complete case analysis for regression models with a right-censored covariate." *The American Statistician* 78.3 (2024): 335-344.
9. Lotspeich, Sarah C., et al. "Making Sense of Censored Covariates: Statistical Methods for Studies of Huntington's Disease." *Annual review of statistics and its application* 11 (2024).
10. Lee, Seong-ho, et al. "SPARCC: Semi-Parametric Robust Estimation in a Right-Censored Covariate Model." *Journal of the American Statistical Association* just-accepted (2025): 1-22.
11. Qian, Jing, et al. "Threshold regression to accommodate a censored covariate." *Biometrics* 74.4 (2018): 1261-1270.
12. Rodrigues, Filipe B., Lori Quinn, and Edward J. Wild. "Huntington's disease clinical trials corner: January 2019." *Journal of Huntington's disease* 8.1 (2019): 115-125.
13. Lv, Xiaofeng, et al. "Maximum weighted likelihood for discrete choice models with a dependently censored covariate." *Journal of the Korean Statistical Society* 46.1 (2017): 15-27.
14. Atem, Folefac D., Roland A. Matsouaka, and Vincent E. Zimmern. "Cox regression model with randomly censored covariates." *Biometrical Journal* 61.4 (2019): 1020-1032.



Recipe for deriving augmented estimators

Let $\boldsymbol{\Psi}$ be the influence function of a consistent m-estimator. We define the efficiency condition as,

$$\begin{aligned} \mathbf{0} &= \text{COV}(\boldsymbol{\Psi}_{\text{AUG}} - \boldsymbol{\Psi}, \boldsymbol{\Psi}_{\text{AUG}}) \\ &= \text{var}(\boldsymbol{\Psi}_{\text{AUG}}) - \text{COV}(\boldsymbol{\Psi}, \boldsymbol{\Psi}_{\text{AUG}}). \end{aligned}$$

Under the efficiency condition we satisfy,

$$\rightarrow \text{var}(\boldsymbol{\Psi}) > \text{var}(\boldsymbol{\Psi}_{\text{AUG}}).$$

Augmented estimator is more efficient than the original estimator.